In Problems 29 and 30, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$.

29. $\mathbf{F}(x, y) = y^3\mathbf{i} - x^2y\mathbf{j}$; $\mathbf{r}(t) = e^{-2t}\mathbf{i} + e^t\mathbf{j}$, $0 \le t \le \ln 2$

30. $\mathbf{F}(x, y, z) = e^x\mathbf{i} + xe^{xy}\mathbf{j} + xye^{xyz}\mathbf{k}$; $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$, $0 \le t \le 1$

31. Find the work done by the force $\mathbf{F}(x, y) = y\mathbf{i} + x\mathbf{j}$ acting along $y = \ln x$ from $(1, 0)$ to $(e, 1)$.

32. Find the work done by the force $\mathbf{F}(x, y) = 2xy\mathbf{i} + 4y^2\mathbf{j}$ acting along the piecewise-smooth curve consisting of the line segments from $(-2, 2)$ to $(0, 0)$ and from $(0, 0)$ to $(2, 3)$.

33. Find the work done by the force $\mathbf{F}(x, y) = (x + 2y)\mathbf{i} + (6y - 2x)\mathbf{j}$ acting counterclockwise once around the triangle with vertices $(1, 1)$, $(3, 1)$, and $(3, 2)$.

34. Find the work done by the force $\mathbf{F}(x, y, z) = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$ acting along the curve given by $\mathbf{r}(t) = t^3\mathbf{i} + t^2\mathbf{j} + t\mathbf{k}$ from $t = 1$ to $t = 3$.

35. Find the work done by a constant force $\mathbf{F}(x, y) = a\mathbf{i} + b\mathbf{j}$ acting counterclockwise once around the circle $x^2 + y^2 = 9$.

36. In an inverse square force field $\mathbf{F} = c\mathbf{r}/\|\mathbf{r}\|^3$, where c is a constant and $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$,* find the work done in moving a particle along the line from $(1, 1, 1)$ to $(3, 3, 3)$.

37. Verify that the line integral $\int_C y^2\,dx + xy\,dy$ has the same value on C for each of the following parameterizations:

$$C: x = 2t + 1, \quad y = 4t + 2, \quad 0 \le t \le 1$$
$$C: x = t^2, \quad y = 2t^2, \quad 1 \le t \le \sqrt{3}$$
$$C: x = \ln t, \quad y = 2\ln t, \quad e \le t \le e^3.$$

38. Consider the three curves between $(0, 0)$ and $(2, 4)$:

$$C_1: x = t, \quad y = 2t, \quad 0 \le t \le 2$$
$$C_2: x = t, \quad y = t^2, \quad 0 \le t \le 2$$
$$C_3: x = 2t - 4, \quad y = 4t - 8, \quad 2 \le t \le 3.$$

Show that $\int_{C_1} xy\,ds = \int_{C_3} xy\,ds$, but $\int_{C_1} xy\,ds \ne \int_{C_2} xy\,ds$. Explain.

39. Assume a smooth curve C is described by the vector function $\mathbf{r}(t)$ for $a \le t \le b$. Let acceleration, velocity, and speed be given by $\mathbf{a} = d\mathbf{v}/dt$, $\mathbf{v} = d\mathbf{r}/dt$, and $v = \|\mathbf{v}\|$, respectively. Using Newton's second law $\mathbf{F} = m\mathbf{a}$, show that, in the absence of friction, the work done by $\mathbf{F}$ in moving a particle of constant mass m from point A at $t = a$ to point B at $t = b$ is the same as the change in kinetic energy:

$$K(B) - K(A) = \frac{1}{2}m[v(b)]^2 - \frac{1}{2}m[v(a)]^2.$$

[*Hint*: Consider $\dfrac{d}{dt}v^2 = \dfrac{d}{dt}\mathbf{v} \cdot \mathbf{v}$.]

40. If $\rho(x, y)$ is the density of a wire (mass per unit length), then $m = \int_C \rho(x, y)\,ds$ is the mass of the wire. Find the mass of a wire having the shape of the semicircle $x = 1 + \cos t$, $y = \sin t$, $0 \le t \le \pi$, if the density at a point P is directly proportional to distance from the y-axis.

41. The coordinates of the center of mass of a wire with variable density are given by $\bar{x} = M_y/m$, $\bar{y} = M_x/m$, where

$$m = \int_C \rho(x, y)\,ds, \quad M_x = \int_C y\,\rho(x, y)\,ds$$

and
$$M_y = \int_C x\,\rho(x, y)\,ds.$$

Find the center of mass of the wire in Problem 40.

42. A force field $\mathbf{F}(x, y)$ acts at each point on the curve C, which is the union of C_1, C_2, and C_3 shown in red in **FIGURE 9.8.21**. $\|\mathbf{F}\|$ is measured in pounds and distance is measured in feet using the scale given in the figure. Use the representative vectors shown to approximate the work done by $\mathbf{F}$ along C. [*Hint*: Use $W = \int_C \mathbf{F} \cdot \mathbf{T}\,ds$.]

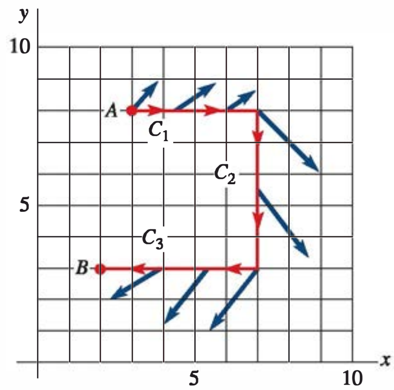


FIGURE 9.8.21 Force field in Problem 42

*Note that the magnitude of $\mathbf{F}$ is inversely proportional to $\|\mathbf{r}\|^2$.

9.9 Independence of the Path

Introduction In this section we refer to a piecewise-smooth curve C between an initial point A and a terminal point B as a **path of integration** or simply a **path**. We begin the discussion by considering line integrals in 2-space. Recall, in (10) of Section 9.8 we saw that if $\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$ is a vector field in 2-space and a path C is defined by the vector function $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$, $a \le t \le b$, then a line integral can be written as

$$\int_C P\,dx + Q\,dy = \int_C \mathbf{F} \cdot d\mathbf{r},$$

where $d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j}$. Generally the value of a line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ depends on the path of integration. Stated another way, if C_1 and C_2 are two different paths between the same points A and B, then we expect that $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} \neq \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$. However, there is an important exception. As we see in this section, line integrals that involve a certain kind of vector field $\mathbf{F}$ depend not on the path C but only on the endpoints $A = (f(a), g(a))$ and $B = (f(b), g(b))$ of the path.

Note: To avoid needless repetition we assume throughout that the component functions of the vector field $\mathbf{F}$ are continuous in some region of 2- or 3-space, its component functions have continuous first partial derivatives in the region, and that the path C of integration lies entirely in the region.

EXAMPLE 1 Path Independence

The integral $\int_C y\,dx + x\,dy$ has the same value on each path C between $(0, 0)$ and $(1, 1)$ shown in **FIGURE 9.9.1**. You may recall from Problems 11–14 in Exercises 9.8 that on these paths

$$\int_C y\,dx + x\,dy = 1.$$

You are also urged to verify $\int_C y\,dx + x\,dy = 1$ on the curves $y = x^3$, $y = x^4$, and $y = \sqrt{x}$ between $(0, 0)$ and between $(1, 1)$. The relevance of all this is to suggest that the integral $\int_C y\,dx + x\,dy$ does not depend on the path joining these two points. We continue this discussion in Examples 2 and 3.

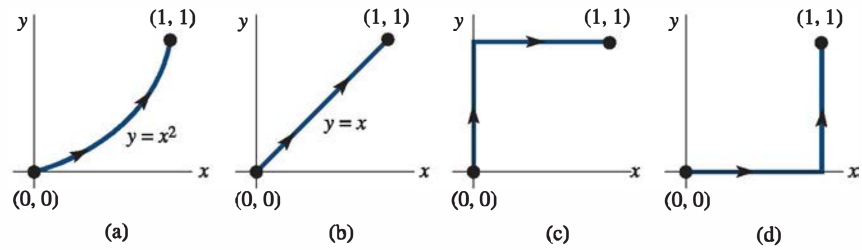


FIGURE 9.9.1 Line integral in Example 1 is the same on four paths ≡

□ Conservative Vector Fields

Before proceeding, we need to introduce a special kind of vector field $\mathbf{F}$ called a **conservative field**.

> **Definition 9.9.1 Conservative Vector Field**
>
> A vector function $\mathbf{F}$ in 2- or 3-space is said to be **conservative** if $\mathbf{F}$ can be written as the gradient of a scalar function ϕ. The function ϕ is called a **potential function** for $\mathbf{F}$.

In other words, $\mathbf{F}$ is conservative if there exists a function ϕ such that $\mathbf{F} = \nabla\phi$. A conservative vector field is also called a **gradient vector field**.

EXAMPLE 2 Conservative Vector Field

The integral in Example 1 can be interpreted as a line integral of a vector field $\mathbf{F}$ along a path C. If $\mathbf{F}(x, y) = y\mathbf{i} + x\mathbf{j}$ and $d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j}$, then $\int_C y\,dx + x\,dy = \int_C \mathbf{F} \cdot d\mathbf{r}$. Now consider the function $\phi(x, y) = xy$. The gradient of the scalar function ϕ is

$$\nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} = y\mathbf{i} + x\mathbf{j}.$$

Because $\nabla\phi = \mathbf{F}(x, y)$ we conclude that $\mathbf{F}(x, y) = y\mathbf{i} + x\mathbf{j}$ is a conservative vector field and that ϕ is a potential function for $\mathbf{F}$. ≡

Of course, not every vector field is a conservative field although many vector fields encountered in physics are conservative. For present purposes, the importance of conservative vector fields will become evident as we continue our study of line integrals.

Path Independence If the value of a line integral is the same for *every* path in a region connecting the initial point A and terminal point B, then the integral is said to be **independent of the path**. In other words, a line integral $\int_C \mathbf{F}\cdot d\mathbf{r}$ of $\mathbf{F}$ along C is independent of the path if $\int_{C_1}\mathbf{F}\cdot d\mathbf{r} = \int_{C_2}\mathbf{F}\cdot d\mathbf{r}$ for any two paths C_1 and C_2 between A and B.

A Fundamental Theorem The next theorem establishes an important relationship between the value of a line integral $\int_C \mathbf{F}\cdot d\mathbf{r}$ and the fact that its path of integration C lies within a conservative vector field $\mathbf{F}$. In addition, it provides a means of evaluating these line integrals in a manner that is akin to the Fundamental Theorem of Calculus:

$$\int_a^b f'(x)\,dx = f(b) - f(a), \tag{1}$$

where $f(x)$ is an antiderivative of $f'(x)$. In the next theorem, known as the **Fundamental Theorem for Line Integrals**, the gradient $\nabla\phi$ of a scalar function ϕ plays the part of the derivative $f'(x)$ in (1).

Theorem 9.9.1 Fundamental Theorem

Suppose C is a path in an open region R of the xy-plane and is defined by $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$, $a \le t \le b$. If $\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$ is a conservative vector field in R and ϕ is a potential function for $\mathbf{F}$, then

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C \nabla\phi\cdot d\mathbf{r} = \phi(B) - \phi(A), \tag{2}$$

where $A = (x(a), y(a))$ and $B = (x(b), y(b))$.

PROOF: We will prove the theorem for a smooth path C. Because ϕ is a potential function for $\mathbf{F}$ we have

$$\mathbf{F} = \nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j}.$$

Then using $\mathbf{r}'(t) = (dx/dt)\mathbf{i} + (dy/dt)\mathbf{j}$ we can write the line integral of $\mathbf{F}$ along the path C as

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C \mathbf{F}\cdot\mathbf{r}'(t)\,dt = \int_a^b \left(\frac{\partial\phi}{\partial x}\frac{dx}{dt} + \frac{\partial\phi}{\partial y}\frac{dy}{dt}\right)dt.$$

In view of the Chain Rule (see (8) in Section 9.4),

$$\frac{d\phi}{dt} = \frac{\partial\phi}{\partial x}\frac{dx}{dt} + \frac{\partial\phi}{\partial y}\frac{dy}{dt}$$

it follows that

$$\begin{aligned}\int_C \mathbf{F}\cdot d\mathbf{r} &= \int_a^b \frac{d\phi}{dt}\,dt = \phi(x(t), y(t))\Big]_a^b \\ &= \phi(x(b), y(b)) - \phi(x(a), y(a)) \\ &= \phi(B) - \phi(A).\end{aligned}$$

≡

For piecewise-smooth curves, the foregoing proof must be modified by considering each smooth arc of the curve C.

Theorem 9.9.1 shows that if $\mathbf{F}$ is a conservative vector field in an open region in 2- or 3-space, then $\int_C \mathbf{F} \cdot d\mathbf{r}$ depends only on the initial and terminal points A and B of the path C, and not on C itself. In other words, line integrals of conservative vector fields are independent of the path. Such integrals are often written

$$\int_A^B \mathbf{F} \cdot d\mathbf{r} = \int_A^B \nabla\phi \cdot d\mathbf{r}. \tag{3}$$

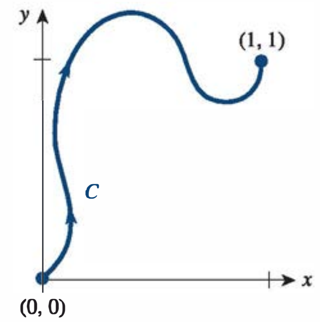


FIGURE 9.9.2 Piecewise-smooth curve in Example 3

EXAMPLE 3 Example 1 Revisited Again

Evaluate $\int_C y\,dx + x\,dy$, where C is a path with initial point (0, 0) and terminal point (1, 1).

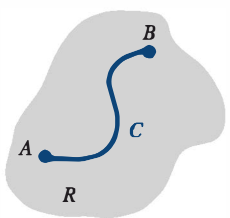
SOLUTION The path C shown in **FIGURE 9.9.2** represents any piecewise-smooth curve with initial and terminal points (0, 0) and (1, 1). We have just seen that $\mathbf{F} = y\mathbf{i} + x\mathbf{j}$ is a conservative vector field defined at each point of the xy-plane and that $\phi(x, y) = xy$ is a potential function for $\mathbf{F}$. Thus, in view of (2) of Theorem 9.9.1 and (3), we can write

$$\begin{aligned}
\int_C y\,dx + x\,dy &= \int_{(0,0)}^{(1,1)} \mathbf{F} \cdot d\mathbf{r} = \int_{(0,0)}^{(1,1)} \nabla\phi \cdot d\mathbf{r} \\
&= xy \Big]_{(0,0)}^{(1,1)} \\
&= 1 \cdot 1 - 0 \cdot 0 = 1.
\end{aligned}$$

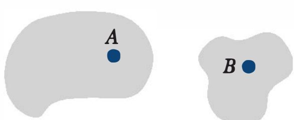
≡

In using the Fundamental Theorem of Calculus (1), *any* antiderivative of $f'(x)$ can be used, such as $f(x) + K$, where K is a constant. Similarly, a potential function for the vector field in Example 2 is $\phi(x, y) = xy + K$ where K is a constant. We may disregard this constant when using (2) of Theorem 9.9.1 since

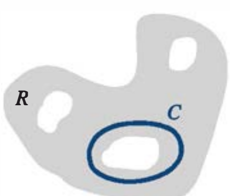
$$\int_A^B \mathbf{F} \cdot d\mathbf{r} = (\phi(B) + K) - (\phi(A) + K) = \phi(B) - \phi(A).$$



(a) Connected region R



(b) R is not connected



(c) Multiply connected region R

FIGURE 9.9.3 Regions in the plane

Some Terminology We say that a region (in the plane or in space) is **connected** if every pair of points A and B in the region can be joined by a piecewise-smooth curve that lies entirely in the region. A region R in the plane is **simply connected** if it is connected and every simple closed curve C lying entirely within the region can be shrunk, or contracted, to a point without leaving R. The last condition means that if C is any simple closed curve lying entirely in R, then the region in the interior of C also lies entirely in R. Roughly put, a simply connected region has no holes in it. The region R in **FIGURE 9.9.3(a)** is a simply connected region. In Figure 9.9.3(b) the region R shown is not connected, or **disconnected**, since A and B cannot be joined by a piecewise-smooth curve C that lies in R. The region in Figure 9.9.3(c) is connected but not simply connected because it has three holes in it. The representative curve C in the figure surrounds one of the holes, and so cannot be shrunk to a point without leaving the region. This last region is said to be **multiply connected**. Lastly, a region R is said to be **open** if it contains no boundary points.

In an open connected region R, the notions of path independence and a conservative vector field are equivalent. This means: If $\mathbf{F}$ is conservative in R, then $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path C, and conversely, if $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path, then $\mathbf{F}$ is conservative.

We state this formally in the next theorem.

Theorem 9.9.2 Equivalent Concepts

In an open connected region R, $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path C if and only if the vector field $\mathbf{F}$ is conservative in R.

PROOF: If $\mathbf{F}$ is conservative in R, then we have already seen that $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path C as a consequence of Theorem 9.9.1.

For convenience we prove the converse for a region R in the plane. Assume that $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path in R and that (x_0, y_0) and (x, y) are arbitrary points in the region R. Let the function $\phi(x, y)$ be defined as

$$\phi(x, y) = \int_{(x_0, y_0)}^{(x, y)} \mathbf{F} \cdot d\mathbf{r},$$

where C is an arbitrary path in R from (x_0, y_0) to (x, y) and $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$. See **FIGURE 9.9.4(a)**. Now choose a point (x_1, y), $x_1 \neq x$, so that the line segment from (x_1, y) to (x, y) is in R. See Figure 9.9.4(b). Then by path independence we can write

$$\phi(x, y) = \int_{(x_0, y_0)}^{(x_1, y)} \mathbf{F} \cdot d\mathbf{r} + \int_{(x_1, y)}^{(x, y)} \mathbf{F} \cdot d\mathbf{r}.$$

Now,

$$\frac{\partial \phi}{\partial x} = 0 + \frac{\partial}{\partial x} \int_{(x_1, y)}^{(x, y)} P\,dx + Q\,dy$$

since the first integral does not depend on x. But on the line segment between (x_1, y) and (x, y), y is constant so that $dy = 0$. Hence, $\int_{(x_1,y)}^{(x,y)} P\,dx + Q\,dy = \int_{(x_1,y)}^{(x,y)} P\,dx$. By the derivative form of the Fundamental Theorem of Calculus we then have

$$\frac{\partial \phi}{\partial x} = \frac{\partial}{\partial x} \int_{(x_1, y)}^{(x, y)} P(x, y)\,dx = P(x, y).$$

Likewise we can show that $\partial\phi/\partial y = Q(x, y)$. Hence, from

$$\nabla \phi = \frac{\partial \phi}{\partial x}\mathbf{i} + \frac{\partial \phi}{\partial y}\mathbf{j} = P\mathbf{i} + Q\mathbf{j} = \mathbf{F}(x, y)$$

we conclude that $\mathbf{F}$ is conservative. ≡

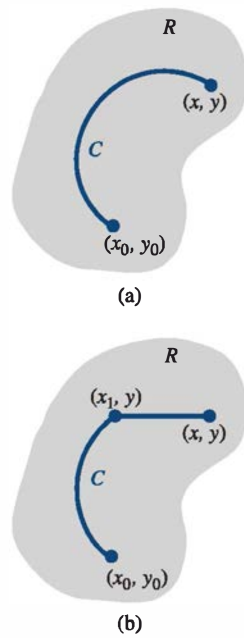


FIGURE 9.9.4 Region R in the proof of Theorem 9.9.2

☐ Integrals Around Closed Paths

Recall from Section 9.8 that a path, or curve, C is said to be closed when its initial point A is the same as the terminal point B. If C is a parametric curve defined by a vector function $\mathbf{r}(t)$, $a \leq t \leq b$, then C is **closed** when $A = B$; that is, $\mathbf{r}(a) = \mathbf{r}(b)$. The next theorem is an immediate consequence of Theorem 9.9.1.

Theorem 9.9.3 Equivalent Concepts

In an open connected region R, $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path if and only if $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path C in R.

PROOF: First, we show that if $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path, then $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path C in R. To see this let us suppose A and B are any two points on C and that $C = C_1 \cup C_2$, where C_1 is a path from A to B and C_2 is a path from B to A. See **FIGURE 9.9.5(a)**. Then

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} - \int_{-C_2} \mathbf{F} \cdot d\mathbf{r}, \tag{4}$$

where $-C_2$ is now a path from A to B. Because of path independence $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{-C_2} \mathbf{F} \cdot d\mathbf{r}$. Thus (4) implies that $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$.

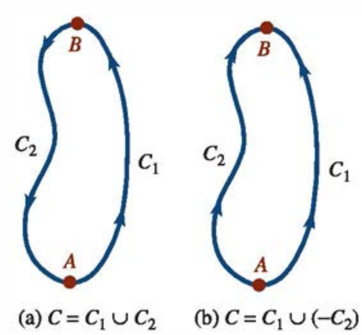


FIGURE 9.9.5 Paths in the proof of Theorem 9.9.3

Next, we prove the converse that if $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path C in R, then $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path. Let C_1 and C_2 represent any two paths from A to B and so $C = C_1 \cup (-C_2)$ is a closed path. See Figure 9.9.5(b). It follows from $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ or

$$0 = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{-C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} - \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$$

that $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$. Hence $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path. ≡

Suppose $\mathbf{F}$ is a conservative vector field defined over an open connected region and C is a closed path lying entirely in the region. When the results of the preceding theorems are put together we conclude that:

$$\mathbf{F} \text{ conservative} \Leftrightarrow \text{path independence} \Leftrightarrow \int_C \mathbf{F} \cdot d\mathbf{r} = 0. \tag{5}$$

The symbol $\Leftrightarrow$ in (5) is read "equivalent to" or "if and only if."

Test for a Conservative Field

The implications in (5) show that if the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is not path independent, then the vector field is not conservative. But there is an easier way of determining whether $\mathbf{F}$ is conservative. The following theorem is a test for a conservative vector field that uses the partial derivatives of the component functions of $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$.

Theorem 9.9.4 Test for a Conservative Field

Suppose $\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$ is a conservative vector field in an open region R, and that P and Q are continuous and have continuous first partial derivatives in R. Then

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \tag{6}$$

for all (x, y) in R. Conversely, if the equality (6) holds for all (x, y) in a simply connected region R, then $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ is conservative in R.

PARTIAL PROOF: We prove the first half of the theorem. Assume that the component functions of the conservative vector field $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ are continuous and have continuous first partial derivatives in an open region R. Since $\mathbf{F}$ is conservative there exists a potential function ϕ such that

$$\mathbf{F} = P\mathbf{i} + Q\mathbf{j} = \nabla\phi = \frac{\partial \phi}{\partial x}\mathbf{i} + \frac{\partial \phi}{\partial y}\mathbf{j}.$$

Thus $P = \partial\phi/\partial x$ and $Q = \partial\phi/\partial y$. Now

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}\left(\frac{\partial \phi}{\partial x}\right) = \frac{\partial^2 \phi}{\partial y \partial x} \quad \text{and} \quad \frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial \phi}{\partial y}\right) = \frac{\partial^2 \phi}{\partial x \partial y}.$$

From continuity of the partial derivatives we have $\partial P/\partial y = \partial Q/\partial x$ as was to be shown. ≡

EXAMPLE 4 Using Theorem 9.9.4

The conservative vector field $\mathbf{F}(x, y) = y\mathbf{i} + x\mathbf{j}$ in Example 2 is continuous and has component functions that have continuous first partial derivatives throughout the open region R consisting of the entire xy-plane. With the identifications $P = y$ and $Q = x$ it follows from (6) of Theorem 9.9.4,

$$\frac{\partial P}{\partial y} = 1 = \frac{\partial Q}{\partial x}.$$

≡

EXAMPLE 5 Using Theorem 9.9.4

Determine whether the vector field $\mathbf{F}(x, y) = (x^2 - 2y^3)\mathbf{i} + (x + 5y)\mathbf{j}$ is conservative.

SOLUTION With $P = x^2 - 2y^3$ and $Q = x + 5y$, we find

$$\frac{\partial P}{\partial y} = -6y^2 \quad \text{and} \quad \frac{\partial Q}{\partial x} = 1.$$

Because $\partial P/\partial y \neq \partial Q/\partial x$ for all points in the plane, it follows from Theorem 9.9.4 that **F** is not conservative. ≡

EXAMPLE 6 Using Theorem 9.9.4

Determine whether the vector field $\mathbf{F}(x, y) = -ye^{-xy}\mathbf{i} - xe^{-xy}\mathbf{j}$ is conservative.

SOLUTION With $P = -ye^{-xy}$ and $Q = -xe^{-xy}$, we find

$$\frac{\partial P}{\partial y} = xye^{-xy} - e^{-xy} = \frac{\partial Q}{\partial x}.$$

The components of **F** are continuous and have continuous partial derivatives. Thus (6) holds throughout the xy-plane, which is a simply connected region. From the converse in Theorem 9.9.4 we can conclude that **F** is conservative. ≡

We have one more important question to answer in this section:

If **F** *is a conservative vector field, how does one find a potential function ϕ for* **F**? (7)

In the next example we give the answer to the question posed in (7).

EXAMPLE 7 Integral That Is Path Independent

(a) Show that $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y) = (y^2 - 6xy + 6)\mathbf{i} + (2xy - 3x^2 - 2y)\mathbf{j}$, is independent of the path C between $(-1, 0)$ and $(3, 4)$.

(b) Find a potential function ϕ for **F**.

(c) Evaluate $\displaystyle\int_{(-1,0)}^{(3,4)} \mathbf{F} \cdot d\mathbf{r}$.

SOLUTION **(a)** Identifying $P = y^2 - 6xy + 6$ and $Q = 2xy - 3x^2 - 2y$ yields

$$\frac{\partial P}{\partial y} = 2y - 6x = \frac{\partial Q}{\partial x}.$$

The vector field **F** is conservative because (6) holds throughout the xy-plane and as a consequence the integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path between any two points A and B in the plane.

(b) Because **F** is conservative there is a potential function ϕ such that

$$\frac{\partial \phi}{\partial x} = y^2 - 6xy + 6 \quad \text{and} \quad \frac{\partial \phi}{\partial y} = 2xy - 3x^2 - 2y. \tag{8}$$

Employing partial integration on the first expression in (8) gives

$$\phi = \int (y^2 - 6xy + 6)\, dx = xy^2 - 3x^2y + 6x + g(y), \tag{9}$$

In partial integration with respect to x, the variable y is treated as a constant. Similarly, in partial integration with respect to y we treat x as a constant.

where $g(y)$ is the "constant" of integration. Now we take the partial derivative of (9) with respect to y and equate it to the second expression in (8):

$$\frac{\partial \phi}{\partial y} = 2xy - 3x^2 + g'(y) = 2xy - 3x^2 - 2y.$$

From the last equality we find $g'(y) = -2y$. Integrating again gives $g(y) = -y^2 + C$, where C is a constant. Thus

$$\phi = xy^2 - 3x^2y + 6x - y^2 + C. \tag{10}$$

(c) We can now use Theorem 9.9.2 and the potential function (10) (without the constant) to evaluate the line integral:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{(-1,0)}^{(3,4)} \mathbf{F} \cdot d\mathbf{r} = (xy^2 - 3x^2y + 6x - y^2)\Big]_{(-1,0)}^{(3,4)}$$

$$= (48 - 108 + 18 - 16) - (-6) = -52.$$

≡

Note: Since the integral in Example 7 was shown to be independent of the path in part (a), we can evaluate it without finding a potential function. We can integrate along ***any convenient curve*** C connecting the given points. In particular, the line $y = x + 1$ is such a curve. Using x as a parameter then gives

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (y^2 - 6xy + 6)\,dx + (2xy - 3x^2 - 2y)\,dy$$

$$= \int_{-1}^{3} [(x+1)^2 - 6x(x+1) + 6]\,dx + [2x(x+1) - 3x^2 - 2(x+1)]\,dx$$

$$= \int_{-1}^{3} (-6x^2 - 4x + 5)\,dx = -52.$$

Conservative Vector Fields in 3-Space

For a three-dimensional conservative vector field

$$\mathbf{F}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$$

and a piecewise-smooth space curve $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, $a \le t \le b$, the basic form of (2) is the same:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla\phi \cdot d\mathbf{r} = \phi(x(b), y(b), z(b)) - \phi(x(a), y(a), z(a)) = \phi(B) - \phi(A). \tag{11}$$

If C is a space curve, a line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path whenever the three-dimensional vector field

$$\mathbf{F}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$$

is conservative. The three-dimensional analogue of Theorem 9.9.4 goes like this: If $\mathbf{F}$ is conservative and P, Q, and R are continuous and have continuous first partial derivatives in some open region of 3-space, then

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}, \quad \frac{\partial P}{\partial z} = \frac{\partial R}{\partial x}, \quad \frac{\partial Q}{\partial z} = \frac{\partial R}{\partial y}. \tag{12}$$

Conversely, if (12) holds throughout an appropriate region of 3-space, then $\mathbf{F}$ is conservative.

The necessity of (12) can be seen from (5) of Section 9.7. If $\mathbf{F}$ is conservative then $\mathbf{F} = \nabla\phi$ and curl $\nabla\phi$ = curl $\mathbf{F} = \mathbf{0}$; that is

$$\text{curl}\,\mathbf{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k} = \mathbf{0}.$$

Setting the three components of the vector curl $\mathbf{F}$ equal to 0 yields (12).

EXAMPLE 8 Integral That Is Path Independent

(a) Show that the line integral

$$\int_C (y + yz)\,dx + (x + 3z^3 + xz)\,dy + (9yz^2 + xy - 1)\,dz$$

is independent of the path C between $(1, 1, 1)$ to $(2, 1, 4)$.

(b) Evaluate $\displaystyle\int_{(1,1,1)}^{(2,1,4)} \mathbf{F}\cdot d\mathbf{r}$.

SOLUTION **(a)** With the identifications

$$\mathbf{F}(x, y, z) = (y + yz)\mathbf{i} + (x + 3z^3 + xz)\mathbf{j} + (9yz^2 + xy - 1)\mathbf{k},$$

$$P = y + yz, \quad Q = x + 3z^3 + xz, \quad \text{and} \quad R = 9yz^2 + xy - 1,$$

we see that the equalities

$$\frac{\partial P}{\partial y} = 1 + z = \frac{\partial Q}{\partial x}, \quad \frac{\partial P}{\partial z} = y = \frac{\partial R}{\partial x}, \quad \text{and} \quad \frac{\partial Q}{\partial z} = 9z^2 + x = \frac{\partial R}{\partial y}$$

hold throughout 3-space. From (12) we conclude that $\mathbf{F}$ is conservative and therefore the integral is independent of the path.

(b) The path C shown in **FIGURE 9.9.6** represents any path with initial and terminal points $(1, 1, 1)$ and $(2, 1, 4)$. To evaluate the integral we again illustrate how to find a potential function $\phi(x, y, z)$ for $\mathbf{F}$ using partial integration.

First we know that

$$\frac{\partial \phi}{\partial x} = P, \quad \frac{\partial \phi}{\partial y} = Q, \quad \text{and} \quad \frac{\partial \phi}{\partial z} = R.$$

Integrating the first of these three equations with respect to x gives

$$\phi = xy + xyz + g(y, z).$$

The derivative of this last expression with respect to y must then be equal to Q:

$$\frac{\partial \phi}{\partial y} = x + xz + \frac{\partial g}{\partial y} = x + 3z^3 + xz.$$

Hence,

$$\frac{\partial g}{\partial y} = 3z^3 \quad \text{implies} \quad g = 3yz^3 + h(z).$$

Consequently,

$$\phi = xy + xyz + 3yz^3 + h(z).$$

The partial derivative of this last expression with respect to z must now be equal to the function R:

$$\frac{\partial \phi}{\partial z} = xy + 9yz^2 + h'(z) = 9yz^2 + xy - 1.$$

From this we get $h'(z) = -1$ and $h(z) = -z + K$. Disregarding K, we can write

$$\phi = xy + xyz + 3yz^3 - z. \tag{13}$$

Finally, from (11) and the potential function (13) we obtain

$$\int_{(1,1,1)}^{(2,1,4)} (y + yz)\,dx + (x + 3z^3 + xz)\,dy + (9yz^2 + xy - 1)\,dz$$

$$= (xy + xyz + 3yz^3 - z)\Big]_{(1,1,1)}^{(2,1,4)} = 198 - 4 = 194. \quad \equiv$$

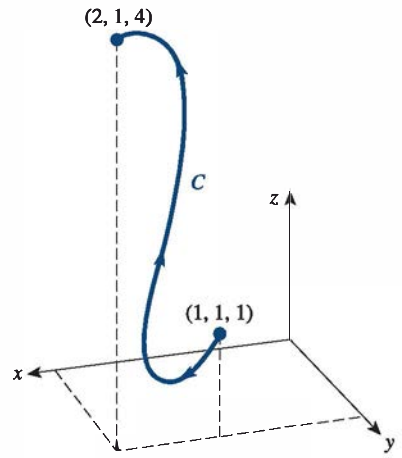


FIGURE 9.9.6 Representative path C in Example 8

Conservation of Energy In a conservative force field $\mathbf{F}$, the work done by the force on a particle moving from position A to position B is the same for all paths between these points. Moreover, the work done by the force along a closed path is *zero*. See Problem 29 in Exercises 9.9. For this reason, such a force field is also said to be **conservative**. In a conservative field $\mathbf{F}$ the **law of conservation of mechanical energy** holds:

For a particle moving along a path in a conservative field,
kinetic energy + potential energy = constant.

See Problem 31 in Exercises 9.9.

A frictional force such as air resistance is **nonconservative**. Nonconservative forces are *dissipative* in that their action reduces kinetic energy without a corresponding increase in potential energy. Stated in another way, if the work done $\int_C \mathbf{F} \cdot d\mathbf{r}$ depends on the path, then $\mathbf{F}$ is nonconservative.

9.9 Exercises

Answers to selected odd-numbered problems begin on page ANS-23.

In Problems 1–10, show that the given line integral is independent of the path. Evaluate in two ways: **(a)** Find a potential function ϕ and then use Theorem 9.9.1, and **(b)** Use any convenient path between the endpoints of the path.

1. $\displaystyle\int_{(0,0)}^{(2,2)} x^2\,dx + y^2\,dy$

2. $\displaystyle\int_{(1,1)}^{(2,4)} 2xy\,dx + x^2\,dy$

3. $\displaystyle\int_{(1,0)}^{(3,2)} (x + 2y)\,dx + (2x - y)\,dy$

4. $\displaystyle\int_{(0,0)}^{(\pi/2,0)} \cos x \cos y\,dx + (1 - \sin x \sin y)\,dy$

5. $\displaystyle\int_{(4,1)}^{(4,4)} \frac{-y\,dx + x\,dy}{y^2}$ on any path not crossing the x-axis

6. $\displaystyle\int_{(1,0)}^{(3,4)} \frac{x\,dx + y\,dy}{\sqrt{x^2 + y^2}}$ on any path not through the origin

7. $\displaystyle\int_{(1,2)}^{(3,6)} (2y^2x - 3)\,dx + (2yx^2 + 4)\,dy$

8. $\displaystyle\int_{(-1,1)}^{(0,0)} (5x + 4y)\,dx + (4x - 8y^3)\,dy$

9. $\displaystyle\int_{(0,0)}^{(2,8)} (y^3 + 3x^2y)\,dx + (x^3 + 3y^2x + 1)\,dy$

10. $\displaystyle\int_{(-2,0)}^{(1,0)} (2x - y \sin xy - 5y^4)\,dx - (20xy^3 + x \sin xy)\,dy$

In Problems 11–16, determine whether the given vector field is a conservative field. If so, find a potential function ϕ for $\mathbf{F}$.

11. $\mathbf{F}(x, y) = (4x^3y^3 + 3)\mathbf{i} + (3x^4y^2 + 1)\mathbf{j}$

12. $\mathbf{F}(x, y) = 2xy^3\mathbf{i} + 3y^2(x^2 + 1)\mathbf{j}$

13. $\mathbf{F}(x, y) = y^2 \cos xy^2\mathbf{i} - 2xy \sin xy^2\mathbf{j}$

14. $\mathbf{F}(x, y) = (x^2 + y^2 + 1)^{-2}(x\mathbf{i} + y\mathbf{j})$

15. $\mathbf{F}(x, y) = (x^3 + y)\mathbf{i} + (x + y^3)\mathbf{j}$

16. $\mathbf{F}(x, y) = 2e^{2y}\mathbf{i} + xe^{2y}\mathbf{j}$

In Problems 17 and 18, find the work done by the force $\mathbf{F}(x, y) = (2x + e^{-y})\mathbf{i} + (4y - xe^{-y})\mathbf{j}$ along the indicated curve.

17.

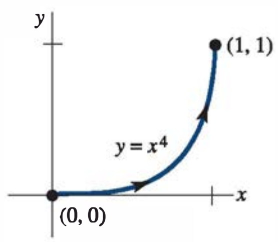


FIGURE 9.9.7 Curve for Problem 17

18.

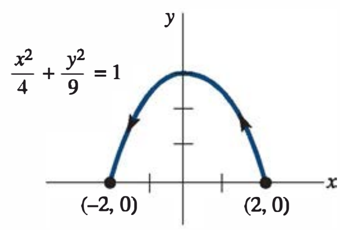


FIGURE 9.9.8 Curve for Problem 18

In Problems 19–24, show that the given integral is independent of the path. Evaluate.

19. $\displaystyle\int_{(1,1,1)}^{(2,4,8)} yz\,dx + xz\,dy + xy\,dz$

20. $\displaystyle\int_{(0,0,0)}^{(1,1,1)} 2x\,dx + 3y^2\,dy + 4z^3\,dz$

21. $\displaystyle\int_{(1,0,0)}^{(2,\pi/2,1)} (2x \sin y + e^{3z})\,dx + x^2 \cos y\,dy + (3xe^{3z} + 5)\,dz$

22. $\displaystyle\int_{(1,2,1)}^{(3,4,1)} (2x + 1)\,dx + 3y^2\,dy + \frac{1}{z}\,dz$

23. $\displaystyle\int_{(1,1,\ln 3)}^{(2,2,\ln 3)} e^{2z}\,dx + 3y^2\,dy + 2xe^{2z}\,dz$